

Notes and Ideas

(parameter-wise functional sensitivity paper)

1 Symmetry Orbits and Infinitesimal Lifting

We study whether the action of a Lie group on a function space can be lifted, at least infinitesimally, through the parameterisation of a neural network. The group- and Lie-algebraic constructions below are standard; the object of interest is the resulting differential lifting problem for the neural-network realisation map.

1.1 Motivation and Setup

Let G be a finite-dimensional Lie group acting smoothly on a manifold $\mathcal{X}$ from the left:

$$\tau : G \times \mathcal{X} \longrightarrow \mathcal{X}, \quad (g, x) \longmapsto \tau_g x.$$

Let (ρ, V) be a finite-dimensional representation of G , and let $\mathcal{B}$ be a G -invariant vector space of differentiable maps $f : \mathcal{X} \rightarrow V$. In function space, the actions on the input and output spaces induce a representation on $\mathcal{B}$:

$$\Pi : G \longrightarrow \mathrm{GL}(\mathcal{B}), \quad [\Pi(g)f](x) := \rho(g)f(\tau_{g^{-1}}x). \quad (1)$$

A function is fixed by Π precisely when it is equivariant with respect to the actions on $\mathcal{X}$ and V :

$$f \in \mathcal{B}^G \iff f(\tau_g x) = \rho(g)f(x) \quad \text{for all } g \in G, x \in \mathcal{X}. \quad (2)$$

This construction describes symmetry directly in function space. In a neural network, however, functions are represented through a finite-dimensional parameter space. We therefore introduce a smooth parameter manifold Θ and a realisation map

$$\Phi : \Theta \longrightarrow \mathcal{B}, \quad \theta \longmapsto f_\theta. \quad (3)$$

The strongest form of the lifting problem asks whether there exists a smooth action

$$\beta : G \times \Theta \longrightarrow \Theta$$

such that

$$\Phi(\beta_g \theta) = \Pi(g)\Phi(\theta)$$

for every $g \in G$ and $\theta \in \Theta$. In other words, we ask whether the realisation map Φ can be made G -equivariant.

Let

$$\mathcal{M} := \Phi(\Theta)$$

denote the realised model class. Although Π defines an action on the ambient function space $\mathcal{B}$, this action need not preserve $\mathcal{M}$. In particular, the transformed function $\Pi(g)f_\theta$ need not be realised by any parameter in Θ . Even when $\mathcal{M}$ is G -invariant, it may not be possible to choose parameter representatives in a way that defines a smooth action on Θ .

Our primary interest is the weaker infinitesimal version of this question. At a fixed parameter value θ , can every first-order variation of f_θ generated by the group action be realised by a first-order variation of the parameters? The following subsections formulate this question by comparing the tangent space to the symmetry orbit of f_θ with the image of the differential of Φ .

1.2 The Differential Lifting Criterion

We now derive a necessary criterion for there to be such a lifting by comparing the infinitesimal symmetry directions in function space with the function variations generated by changes in the network parameters. At $\theta \in \Theta$, the differential of the realisation map is

$$J_\theta := d\Phi_\theta : T_\theta \Theta \longrightarrow T_{f_\theta} \mathcal{B} \cong \mathcal{B}. \quad (4)$$

Thus $J_\theta v$ is the first-order change in the realised function produced by the parameter velocity $v \in T_\theta \Theta$.

When $\Theta \subseteq \mathbb{R}^p$, this differential takes the form

$$[J_\theta v](x) = \sum_{i=1}^p v_i S_i(x; \theta), \quad S_i(x; \theta) := \frac{\partial f_\theta(x)}{\partial \theta_i}. \quad (5)$$

The space of first-order function variations available through the parameterisation is therefore

$$\mathcal{T}_\theta := \text{Im}(J_\theta) = \text{span}\{S_1(\cdot; \theta), \dots, S_p(\cdot; \theta)\}. \quad (6)$$

At a regular point of Φ , this is the tangent space $T_{f_\theta}\mathcal{M}$ of the realised model class.

Suppose now that a smooth parameter action

$$\beta : G \times \Theta \longrightarrow \Theta$$

realises the function-space action. For fixed θ , define the parameter-space orbit map

$$b_\theta : G \longrightarrow \Theta, \quad b_\theta(g) := \beta_g \theta.$$

The equivariance condition

$$\Phi(\beta_g \theta) = \Pi(g)\Phi(\theta)$$

can then be written as the identity

$$\Phi \circ b_\theta = o_{f_\theta}, \quad (7)$$

where $o_{f_\theta}(g) = \Pi(g)f_\theta$ is the function-space orbit map.

Differentiating (7) at the identity $e \in G$ and applying the chain rule gives

$$d\Phi_\theta \circ d(b_\theta)_e = d(o_{f_\theta})_e. \quad (8)$$

Define

$$L_\theta := d(b_\theta)_e : \mathfrak{g} \longrightarrow T_\theta \Theta.$$

For $\xi \in \mathfrak{g}$, $L_\theta(\xi)$ is the parameter velocity generated by ξ . Using the definitions $J_\theta = d\Phi_\theta$ and $a_{f_\theta} = d(o_{f_\theta})_e$, equation (8) becomes

$$J_\theta L_\theta = a_{f_\theta}. \quad (9)$$

Equivalently, for every $\xi \in \mathfrak{g}$,

$$J_\theta L_\theta(\xi) = a_{f_\theta}(\xi) = d\Pi(\xi)f_\theta. \quad (10)$$

Equation (9) immediately yields the necessary condition

$$\text{Im}(a_{f_\theta}) \subseteq \text{Im}(J_\theta). \quad (11)$$

Using

$$\text{Im}(a_{f_\theta}) = T_{f_\theta}(G \cdot f_\theta), \quad \text{Im}(J_\theta) = \mathcal{T}_\theta,$$

this condition can equivalently be written as

$$T_{f_\theta}(G \cdot f_\theta) \subseteq \mathcal{T}_\theta. \quad (12)$$

Thus, if a smooth parameter action realising Π exists, every infinitesimal function-space orbit direction must be attainable through an infinitesimal change of the parameters.

Conversely, at a fixed parameter value θ , the inclusion (11) is sufficient for the existence of some linear map

$$L_\theta : \mathfrak{g} \longrightarrow T_\theta \Theta$$

satisfying (9). It is therefore necessary and sufficient for pointwise first-order lifting. It is not sufficient for the existence of a local or global action on Θ : such an action requires the lifts to vary smoothly with θ , satisfy the appropriate Lie-bracket relations, and obey the relevant integrability conditions.

1.3 A Least-Squares Lifting Objective

The differential lifting condition requires the operator equation

$$J_\theta L_\theta = a_{f_\theta}$$

to have a solution. When this equation cannot be solved exactly, its failure can be quantified through a least-squares relaxation.

Assume that $\mathcal{B}$ is equipped with a Hilbert-space inner product, and choose inner products on $\mathfrak{g}$ and $T_\theta \Theta$. For a candidate linear lift

$$L : \mathfrak{g} \longrightarrow T_\theta \Theta,$$

define the lifting error

$$\mathcal{E}_\theta(L) := \|J_\theta L - a_{f_\theta}\|_{\text{HS}}^2. \quad (13)$$

If $\{\xi_a\}_{a=1}^d$ is an orthonormal basis of $\mathfrak{g}$, then

$$\mathcal{E}_\theta(L) = \sum_{a=1}^d \|J_\theta L(\xi_a) - a_{f_\theta}(\xi_a)\|_{\mathcal{B}}^2. \quad (14)$$

Let P_θ denote the orthogonal projector onto $\mathcal{T}_\theta = \text{Im}(J_\theta)$. Since $J_\theta L$ takes values in $\mathcal{T}_\theta$, we have the orthogonal decomposition

$$\mathcal{E}_\theta(L) = \|J_\theta L - P_\theta a_{f_\theta}\|_{\text{HS}}^2 + \|(I - P_\theta)a_{f_\theta}\|_{\text{HS}}^2. \quad (15)$$

The second term is independent of L . Hence

$$\min_L \mathcal{E}_\theta(L) = \|(I - P_\theta)a_{f_\theta}\|_{\text{HS}}^2. \quad (16)$$

in function space.